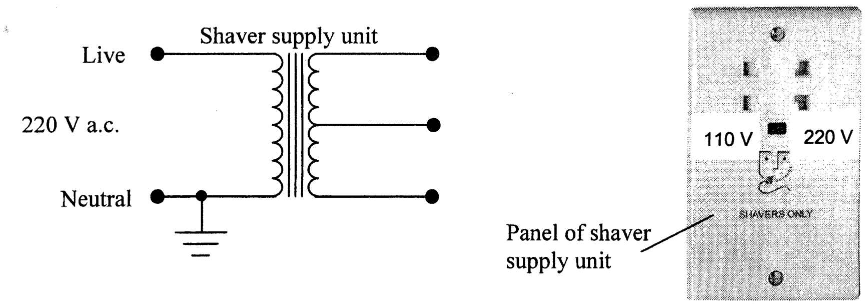
9. Read the following description about the 'shaver supply unit' in bathrooms and answer the questions that follow.

> The danger of electric shock is particularly high in bathrooms. Normal electric socket outlets should not be installed in bathrooms. As electric shavers and toothbrushes are becoming popular these days, a special unit, called 'shaver supply unit' is now common in bathrooms to provide electricity just for these low power consumption electric appliances (Figure 9.1).
>
> The shaver supply unit consists of a transformer in which the secondary is not earthed and is completely isolated from the 220 V a.c. mains supply connecting to the primary. It can be used with 220 V or 110 V shavers.

Figure 9.1



(a) Explain why the chance of electric shock is high in bathrooms. (2 marks)

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(b) Explain what would happen if the human body touches

(i) the live wire of the mains supply in the primary circuit; (2 marks)

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(ii) one of the conducting wires in the shaver circuit outlet. (2 marks)

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*(c) What is the turns ratio of the primary coil to the secondary coil of the transformer so as to provide 110 V ? (1 mark)

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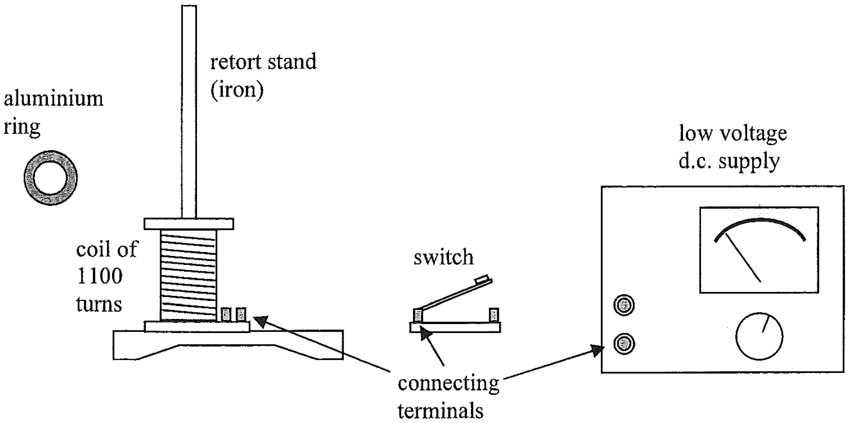
9. (a) You are given a low voltage d.c. supply, an aluminum ring, a switch, a coil of 1100 turns and a retort stand arranged as shown. Use three connecting leads to complete the connections among the apparatus in the figure and describe how to demonstrate Lenz's law in electromagnetic induction. State and explain the observation. (6 marks)



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(b) Describe what would be observed if the experiment in part (a) is repeated with

(i) a low voltage a.c. supply; (1 mark)

(ii) a low voltage a.c. supply and an aluminium ring with a slit cut through it as shown []. (1 mark)

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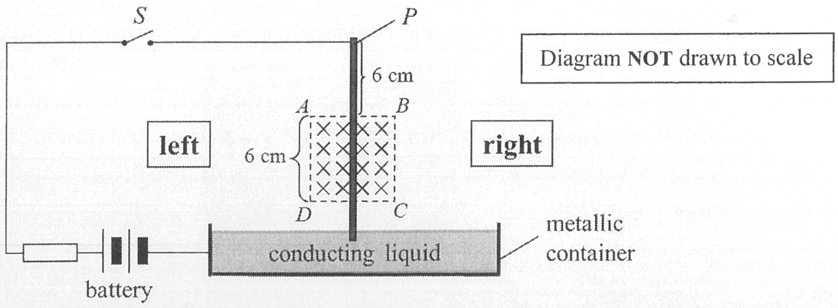
9. Figure 9.1 shows a set-up for demonstrating one of Faraday's discoveries. A light metal rod is free to rotate about point P while its lower end just touches some conducting liquid in a metallic container.

Figure 9.1



A uniform magnetic field pointing into the paper is applied over the region $ABCD$ containing part of the rod. When switch S is closed, the rod 'kicks' out and leaves the liquid surface.

(a) State the direction (to the left / to the right / into the paper / out of the paper) that the rod 'kicks' and describe the subsequent motion of the rod. (3 marks)

(b) When switch S is closed, the initial moment about point P that makes the rod 'kick' out is 7.2×10^{-4} N m. Assume that the magnetic force acts at the midpoint of the part of the rod within the magnetic field.

(i) Calculate the magnetic force acting on the rod at that instant. (2 marks)

(ii) Hence, find the strength B of the magnetic field if the current flowing through the rod is 3.2 A when the circuit is closed. (2 marks)

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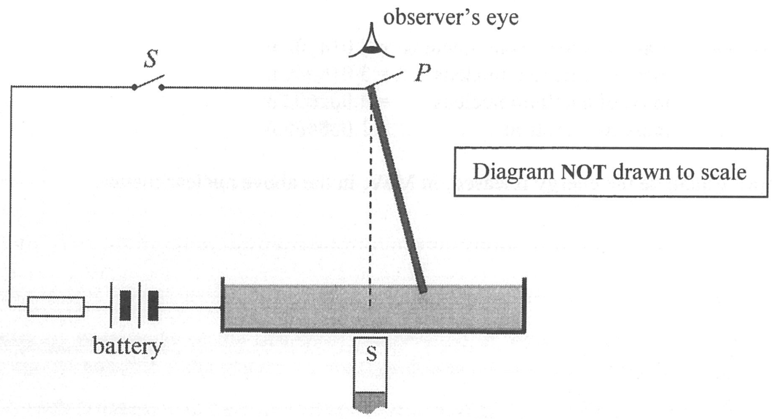
(c) Now the uniform magnetic field is removed and a bar magnet is placed underneath the container as shown in Figure 9.2. The rod is held tilted at an angle to the vertical but with its lower end still in the conducting liquid.

Figure 9.2



(i) Sketch on Figure 9.2 the field lines around the rod due to the bar magnet. (1 mark)

(ii) After closing switch S and the rod is released from rest, describe its subsequent motion viewed from above. (1 mark)

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9. (a)

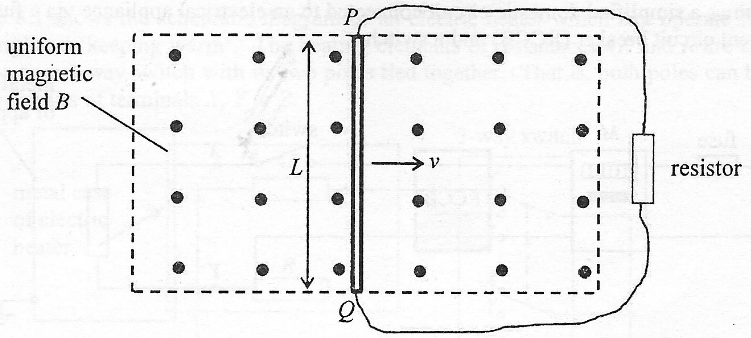

Figure 9.1

Figure 9.1 shows a metal rod PQ of length L moving with constant velocity v across a uniform magnetic field B pointing out of the paper. An e.m.f. ξ is induced across rod PQ as it cuts the field lines. When the rod is connected to a resistor outside the field, a current I flows in the circuit.

(i) Indicate the direction of I in Figure 9.1. (1 mark)

(ii) Explain why an external force F is required to maintain the uniform motion of rod PQ. Find F in terms of the physical quantities given. (3 marks)

(iii) This set-up works as a generator. By considering the mechanical power input by external force F to the set-up, show that $\xi = BLv$. (2 marks)

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(b) At a certain place the Earth's magnetic field runs along the S-N direction such that the field lines make an angle θ with the horizontal as shown in Figure 9.2(a).

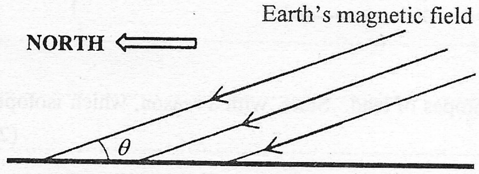

Figure 9.2(a)

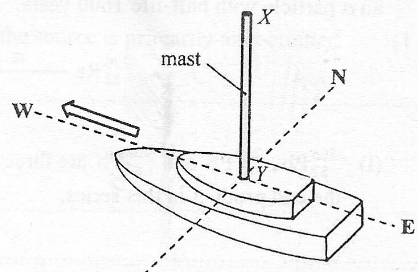

Figure 9.2(b)

A ship with a vertical aluminium mast sails at sea along a straight course to the west as shown in Figure 9.2(b). As a result, an e.m.f. is induced across the mast XY.

(i) Explain why it is **only the horizontal component** of the Earth's magnetic field that is cut by the mast which gives rise to this induced e.m.f. (1 mark)

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(ii) Given: length of mast $XY = 20$ m
speed of the ship = 6 m s^{-1}
Earth's magnetic field = 50 μT
$\theta = 30°$

Referring to (a)(iii), calculate the e.m.f. induced across XY and state whether the distribution of free electrons along the mast is more at end X, more at end Y or uniform along XY. (3 marks)

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(iii) Suppose X and Y are connected by a cable running side-by-side with the mast so that they form a complete circuit. Explain whether there will be a current passing through it. (2 marks)

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7. Read the following passage about **eddy currents** and answer the questions that follow.

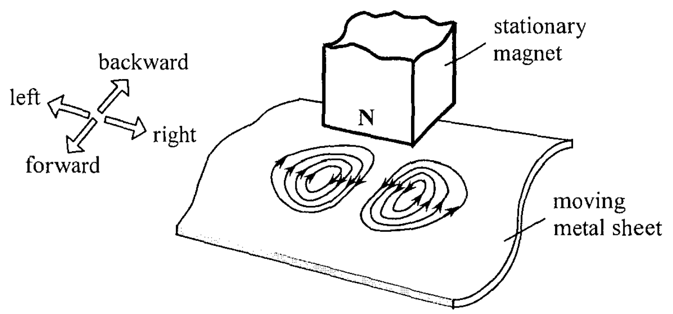
Eddy currents are induced by changing magnetic fields. They flow in closed loops in conductors like swirling eddies in a stream, perpendicular to the direction of the magnetic field. They are commonly applied in braking known as 'eddy braking'.

The heating effect of eddy currents is used in induction heating devices, such as induction cookers. The resistance felt by the eddy currents in a conductor causes Joule heating. However, for applications like motors and transformers, this heat is considered as a waste of energy and as such, eddy currents need to be minimized.

Eddy currents can be removed by cracks or slits in the conductor, which prevent the current loops from circulating. This means that eddy currents can be used in detecting defects in materials. The magnetic field produced by the eddy currents is measured, where a change in the field reveals the presence of an irregularity in the material.

(a) (i) In Figure 7.1, a permanent magnet with north pole facing downwards is held stationary. A metal sheet moves past the magnet (the direction of movement is not shown) and eddy currents are induced as shown. Briefly explain why eddy currents are induced and state whether the metal sheet is moving forward, backward, towards the left or towards the right. (2 marks)

Figure 7.1



(ii) State the energy changes in the process in which the metal sheet is slowing down to stop. (2 marks)

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(iii) Although eddy braking has the advantage of being contactless, traditional frictional braking cannot be totally replaced by eddy braking. Why ? (1 mark)

(b) An induction cooker of rating '220 V, 2000 W' operates for 15 minutes. How much should be paid if 1 kW h of electrical energy costs $1.1 ? (2 marks)

(c) State a method to minimize eddy currents produced in the iron cores of motors and transformers. (1 mark)

(d) Eddy currents can be used to detect defects in materials. When there is a crack in a material, how would the magnetic field due to eddy currents change ? Explain briefly. (2 marks)

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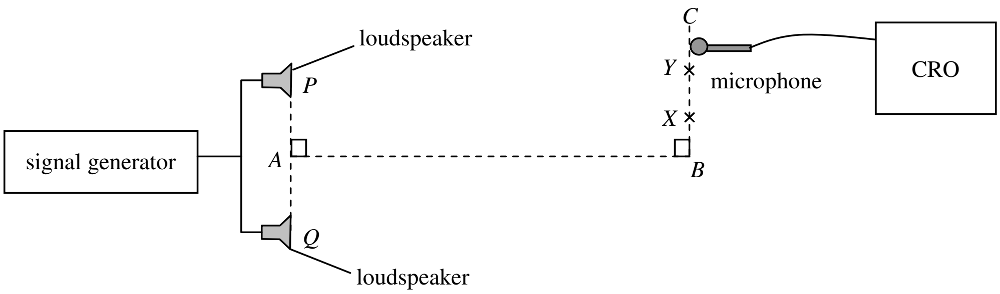

Figure 6.1

Figure 6.1 shows two identical loudspeakers *P* and *Q* are connected to a signal generator. Position *A* is the mid-point of *PQ*. A microphone connected to a CRO is moved along *BC*. The amplitude of the CRO trace increases as the loudness of the sound detected increases. Figure 6.2 shows how the amplitude of the CRO trace varies with the position of the microphone.

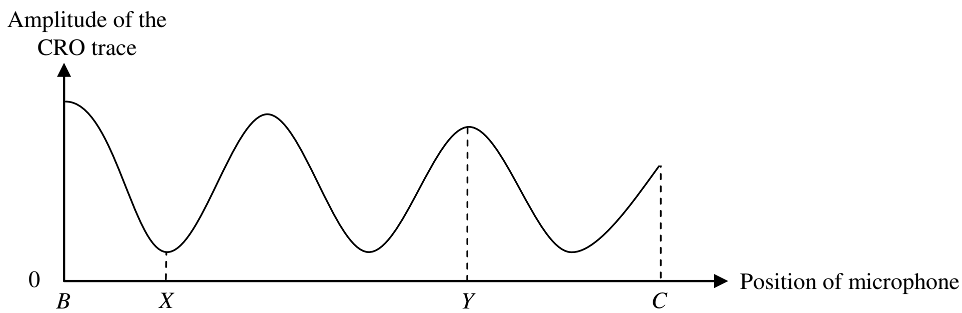

Figure 6.2

(a) (i) Explain why the loudness of the sound varies along *BC*. (2 marks)

(ii) State **ONE** reason why the amplitude of the CRO trace is **NOT** zero at position *X*. (1 mark)

(b) If $PY = 5.10$ m and $QY = 5.78$ m, find the wavelength of the sound. (2 marks)

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7. Amy uses the motor of a toy fan as a simple generator. She connects a bulb to the two terminals of the motor. This is shown in Figure 7.1.

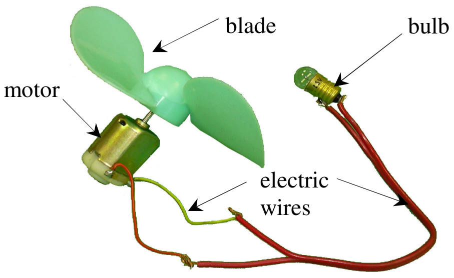

Figure 7.1

The bulb lights up when the blades are turned rapidly. Explain why and state the energy conversion taking place in this process. (4 marks)

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12.

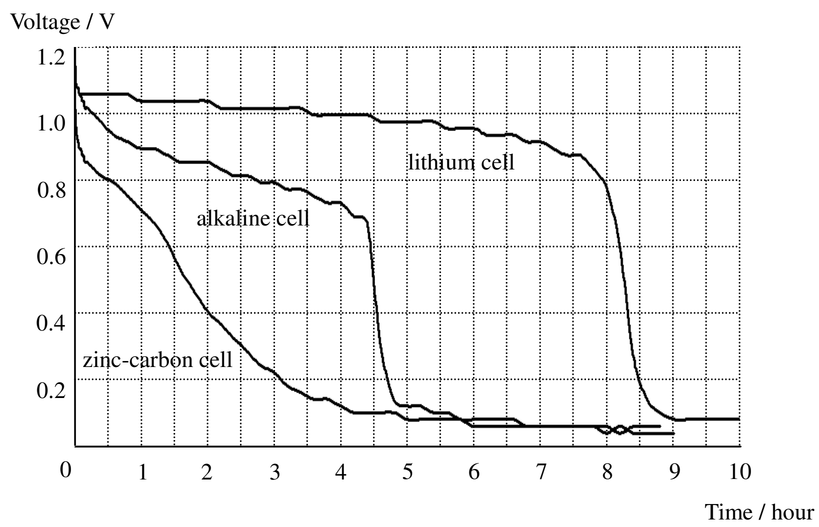
Figure 12.1

Iris uses the apparatus shown in Figure 12.1 to study the lifetime of AA-size cells when used to power a bulb. She connects a cell and a switch to the bulb and uses a voltage sensor to measure the voltage across the bulb.

(a) Draw a circuit diagram to illustrate how the apparatus is connected. Use the symbol (V) to denote the voltage sensor and the data-logger. (2 marks)

(b) Iris conducts the experiment with a zinc-carbon cell, an alkaline cell and a lithium cell separately. Figure 12.2 shows the variation of the voltage across the bulb with time for the cells. The bulb lights up as long as the voltage across it is above 0.6 V.

Figure 12.2



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(b) (i) A salesman claims that the lifetime of a lithium cell for lighting up the bulb is five times that of an alkaline cell. Determine whether the claim is correct or not. (2 marks)

(ii) Table 12.3 shows the prices of the three types of cell.

Type of cells	Price per cell
zinc-carbon	$ 1.5
alkaline	$ 3.8
lithium	$25.0

Table 12.3

Which type of cells is the best buy, in terms of the cost per hour for lighting up the bulb ? Show your calculations. (3 marks)

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13. Josephine conducts an investigation on transformers. Primary and secondary coils are wound on two soft-iron C-cores to form a transformer. She sets up a circuit as shown in Figure 13.1.

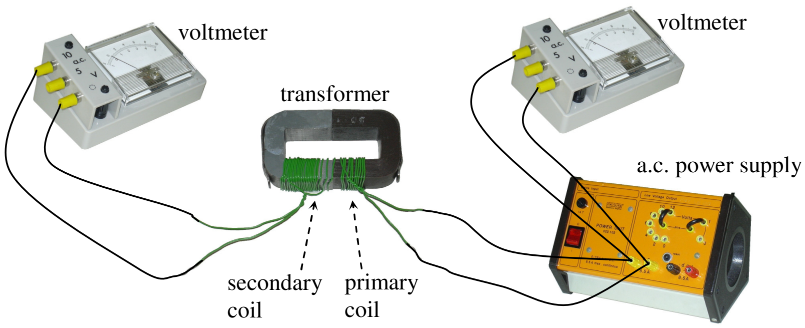

Figure 13.1

*(a) Josephine varies the input voltage V_1 to the transformer and records the corresponding output voltage V_2. The results are shown in Table 13.2. Figure 13.3 shows the graph of V_2 against V_1. Draw a conclusion for this investigation.

V_1 / V	V_2 / V
1.5	2.5
3.0	5.1
4.5	7.6
6.0	10.0

Table 13.2

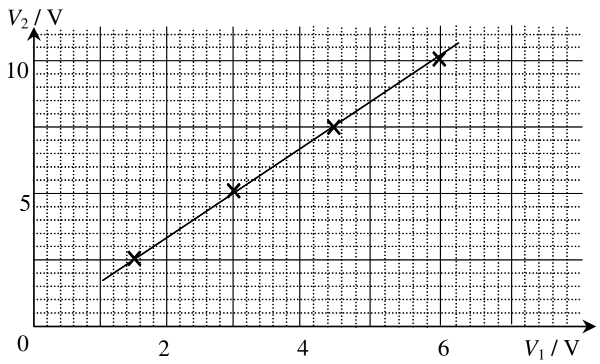

Figure 13.3

(1 mark)

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*(b) Deduce the value of V_2 that will be produced when V_1 equals 8.0 V. (1 mark)

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*(c) Josephine wants to study the relationship between the output voltage and the number of turns in the secondary coil of the transformer. Describe how she can conduct the experiment. (2 marks)

*(d) Josephine adds a bulb to the circuit as shown in Figure 13.4. Suggest how Josephine can estimate the efficiency of the transformer. State the measurement(s) she must take. Additional apparatus may be used if necessary. (3 marks)

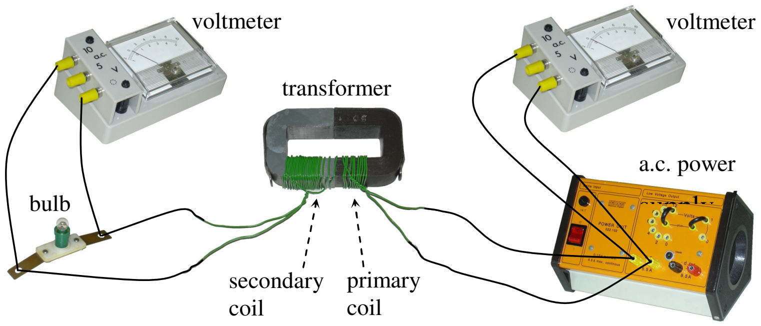

Figure 13.4

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14. In April 1986, a disastrous nuclear accident happened at the Chernobyl Nuclear Power Station. A large quantity of various radioactive substances was released and spread to neighbouring countries. The radiation levels recorded in these countries were much higher than the normal background count rate.

(a) State **ONE** source of background radiation. (1 mark)

(b) One of the radioactive isotopes released in the accident was caesium-137 (Cs-137). The following equation shows how Cs-137 is produced :

$$^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{137}_{55}\mathrm{Cs} + {}^{95}_{37}\mathrm{Rb} + x\,{}^{1}_{0}\mathrm{n}$$

Given : mass of one nuclide of $^{235}_{92}\mathrm{U}$ = 235.0439 u

$^{137}_{55}\mathrm{Cs}$ = 136.9071 u

$^{95}_{37}\mathrm{Rb}$ = 94.9399 u

$^{1}_{0}\mathrm{n}$ = 1.0087 u

1 u is equivalent to 931 MeV

(i) What is the value of x ? (1 mark)

*(ii) Find the energy release in the fission of one U-235 nuclide in MeV. (2 marks)

*(iii) The half-life of Cs-137 is 30 years. A soil sample contaminated by Cs-137 has an activity of 1.2×10^6 Bq (disintegrations per second). A physicist comments that the contaminated sample will affect the environment for more than 350 years. Justify the physicist's claim with calculations. It is known that the activity of an uncontaminated soil sample is 200 Bq. (2 marks)

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